

Ruşen Birben

AI Researcher & Engineer

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AI and Data Engineering graduate from Istanbul Technical University and independent researcher with METU's Applied NLP Group. Research interests center on LLM robustness and reliability, with experience in Turkish NLP, machine unlearning, and LLM applications for finance.

01 🎓 EDUCATION

B.Sc. AI and Data Engineering

2022 – 2025

Istanbul Technical University

GPA: 3.61 / 4.00

Undergraduate studies in Electronics & Communication Engineering

2020 – 2022

Istanbul Technical University

Transferred to AI & Data Engineering

02 🏆 SCHOLARSHIPS & AWARDS

Kaggle Bronze Medal · The Pokémon Company PTCG AI Battle Challenge

2026

Ranked 677th of 6,807 teams (top 10%) in Kaggle's simulation competition, developing and evaluating partially observable game-playing agents with replay-based behavioral cloning, belief-aware Monte Carlo planning, and self-play.

1416 YLSY Scholarship · M.Sc. & Ph.D.

2026

Selected for Türkiye's competitive government scholarship for overseas M.Sc. and Ph.D. study. The 1416 program funds tuition, monthly living support, health and travel costs, language-exam fees, and study supplies in return for a post-degree public-service obligation.

03 🔬 RESEARCH EXPERIENCE

Independent Researcher

03/2026 – Present

METU Applied NLP Group

Ankara, TR

Contribute to data creation and evaluation methodology for the SIGTURK 2027 benchmark on Turkic languages and cultural understanding with Çağrı Toraman. Support LlamaTurk2 through training-data collection and pretraining experiments. Also lead METU NLP Connect, producing expert interviews.

AIOps Research Intern

08/2024 – 09/2024

Havelsan

Istanbul, TR

Investigated software-log analysis for predictive maintenance and developed prototypes using HDFS logs.

NLP Research Intern

07/2024 – 08/2024

ITU NLP Lab

Istanbul, TR

Researched LLM applications in finance, including sentiment analysis, tool use, reasoning, forecasting, and retrieval-augmented generation. Conducted part of the experiments that contributed to Kaira, my undergraduate capstone on financial reasoning with LLMs.

04 PROFESSIONAL EXPERIENCE

Co-Founder & Technical Lead

08/2025 - 04/2026

Fiction Studios 

Ankara, TR

Co-founded Fiction Studios with a friend at Bilkent Cyberpark and served as technical lead. Directed technical development and coordinated three additional contributors supporting the two founders. Built a multilingual LLM-powered guide for museums and tourists, working on the conversational visitor experience alongside customer and investor discussions.

Junior AI Engineer

03/2025 - 07/2025

CyberQuote 

Istanbul, TR

Joined CyberQuote, a Phillip Capital subsidiary, as the second member of its newly formed AI team. Developed news-sentiment pipelines and internal LLM tools for financial professionals, alongside web integrations and a mobile voice assistant. Worked with the practical challenge of retrieving current financial context from fragmented sources and PDF reports, where missing information could lead to unreliable answers.

05 PUBLICATIONS & RESEARCH CONTRIBUTIONS

Turkish Cultural Knowledge Dataset - MRL 2026 Shared Task 2026
Accepted contribution · Joint benchmark paper in preparation [Shared Task](#)
Led the source-grounded agent pipeline and data preparation for an accepted Turkish cultural knowledge dataset spanning 15 domains. Screened 1,200 candidate question-answer pairs, selecting 500 for sociological review by Büşra Tufan and producing a final submission of 300 pairs. Contributed to English translation review, with research guidance from Çağrı Toraman.
My accepted Turkish dataset contribution will form part of the joint multilingual and multimodal benchmark, whose paper I will co-author. The joint dataset and preprint are planned for release in October 2026, with shared task results scheduled for presentation at MRL @ EMNLP 2026. The organizers plan to submit the paper to a high-impact AI venue in 2027.

06 SELECTED PROJECTS

Kaira 2025
Undergraduate Capstone: Financial Reasoning LLM
Explored financial reasoning in Qwen-based LLMs through GRPO fine-tuning with Unsloth and LoRA. Ran FinQA and GSM8K experiments to examine reward design and reasoning behavior, and developed an Ollama-based chat application. Undergraduate capstone at Istanbul Technical University.

AIZheimer 2024
Machine Unlearning in Stable Diffusion [Code](#)
Building on Forget-Me-Not, implemented and adapted cross-attention-based unlearning for Stable Diffusion 2.1. Developed attention controller/processor code and explored attention re-steering, distraction, and random-stimulus loss objectives. Ran qualitative experiments and prepared visualizations to examine the trade-off between removing a target concept and retaining unrelated generation capabilities.

Biting The Bytes 2024
Turkish Diacritic Restoration [Code](#) [Demo](#)
Fine-tuned ByT5-small with PEFT/LoRA for Turkish diacritic restoration, treating the task as byte-level token classification. Took primary responsibility for model training and hyperparameter tuning; shared debugging and Hugging Face Spaces deployment with a teammate. The project explored restoring characters such as ş, ç, ı, ğ, ü, and ö in noisy or ASCII-only Turkish text.

To-AI-or-Not-to-AI 2023
GPT Detector [Code](#) [Demo](#)
Fine-tuned three RoBERTa-based detectors and combined their predictions through majority voting. Shared data creation and preprocessing for a 26,270-example dataset, and deployed the application on Hugging Face Spaces. The ensemble achieved 94.21% accuracy on an in-domain random test split; this evaluation did not establish performance across languages or substantially different domains.

Tobor 2024 - 2025
Autonomous Warehouse Robot [Code](#) [Video](#)
Contributed to a team-built autonomous warehouse robot combining navigation and YOLO-based perception in ROS2. Implemented the box pickup and drop-off mechanism, shared data annotation with the team, and supported work on other algorithms.

07 TECHNICAL SKILLS

Languages Python / JavaScript/TypeScript / SQL / C/C++

Frameworks & Tools PyTorch / TensorFlow / Transformers / PEFT / LoRA / Unsloth / Scikit-learn / Pandas / NumPy / ROS2 / YOLO

Specialties NLP / LLMs / RAG / Machine Unlearning / Model Evaluation

08 COURSES / LANGUAGES

Deep Learning Specialization [DeepLearning.AI · Coursera](#)
Neural networks, CNNs, sequence models, optimization, and structuring machine learning projects.

AWS Cloud Technical Essentials [AWS · Coursera](#)
Cloud architecture and AWS services including EC2, Lambda, S3, RDS, and DynamoDB.

 **Turkish** Native **English** TOEFL iBT 109/120 (November 8, 2025)

09 INTERESTS [rusenbirben.com](#)

Coffee	Books	Photography	Anime & Manga
V60 / Aeropress / Moka Pot	Philosophy / AI/CS / Finance	Street Photography / Double Exposure / Bird Photography	Watcher / Reader / Collector